BD Academy

Training Recommendation System

Prototype v0.8 | Test Cases + Status Report

Course	BD Academy — Training Recommendation System
Submission	Prototype v0.8 — Day 3 Progress
Pipeline status	End-to-end functional (TF-IDF mode) / SBERT (requires internet 1st run)
Datasets	BD_Dataset1_Training_Modules_v3.csv | BD_Dataset2_HCP_Profiles_v3.csv
Test cases	5 documented (TC-01 through TC-05)

Section 1 — Status Note

What works

- End-to-end pipeline runs without errors from raw CSV to ranked recommendation output.
- TF-IDF + LSA embeddings are fully operational vocabulary fitted on module descriptions, customer profile sentences projected into the same space.
- Score fusion (35% TF-IDF + 65% SBERT) with 7 rule-based boosters (language, domain alignment, OPCO, skill, contract, format, duration penalty) is producing differentiated, interpretable scores.
- Category-aware domain boost (replacing clustering) correctly routes customers with specific objectives to the right Topic_Domain in the majority of cases.
- MMR re-ranking ensures diversity across the 5 recommendations the pipeline does not return 5 variants of the same module.

What remains fragile

- This pipeline has been built and tested entirely on synthetic data. While the data was designed to reflect the real BD Academy catalogue structure, it may not fully capture the complexity and edge cases present in production data. Results and observed behaviours should be interpreted with that limitation in mind.
- Sentence-BERT requires an internet connection the first time it runs, in order to download the pre-trained model weights. After that initial download the model is cached locally and the pipeline runs fully offline. The TF-IDF fallback mode handles environments where this first-time download is not yet possible.

What remains to be built

- Cold-start handling for brand-new customers with empty or minimal profiles.
- A lightweight API wrapper so the pipeline can be called programmatically from the BD app.

Section 2 — Five Documented Test Cases

Each test case documents the input customer profile, the 5 recommendations produced by the pipeline, and a colour-coded verdict. A single overall analysis follows at the end of this section.

TC-01 — French Physician — PICC Line Specialist (OPCO Eligible)

VERDICT: PASS

Customer Profile

Customer ID	HCP0079
Role	Physician
Specialty	Cardiac ICU
Country	France
Language	Français (fr) - French
Skill Level	Intermediate
Establishment	MSP (Maison de Santé Pluridisciplinaire)
OPCO Eligible	Yes
Has BD Contract	Yes
Format Preference	On-site
Available Hours	16 hours
Training Objective	Développer les compétences en pose et entretien de PICC line
Preferred Product	BD Alaris™

Pipeline Recommendations

#	ID	Module Title	Topic Domain	Skill Level	Format	Score
1	BD9407	FR_BD Provena™ Midline	Vascular Access & Infusion	Advanced	On-site	**0.6350**
2	BD1724	FR_Secure injection_V1.0	Vascular Access & Infusion	Intermediate	On-site	**0.6186**
3	BD6691	BD Rowa™ Pickup - Recurrent orders	Medication Management	Intermediate	Webinar	**0.4730**
4	BD1810	BD Rowa™ Vmax 160 - Entrée manuelle	Medication Management	Intermediate	On-site	**0.4742**
5	BD4272	ES_Prevención de Oclusiones	Vascular Access & Infusion	Advanced	E-learning	**0.4790**

Score colour: green ≥ 0.65 (strong) | white = moderate | amber < 0.45 (weak)

TC-02 — German Pharmacist — Pyxis ES Optimisation

VERDICT: PASS — BEST CASE

Customer Profile

Customer ID	HCP0077
Role	Pharmacist
Specialty	Cardiac ICU
Country	Germany
Language	Deutsch (de) - German
Skill Level	Intermediate
Establishment	Tagesklinik
OPCO Eligible	No
Has BD Contract	Yes
Format Preference	E-learning
Available Hours	16 hours
Training Objective	Optimise Pyxis ES dispensing workflows
Preferred Product	BD Alaris™

Pipeline Recommendations

#	ID	Module Title	Topic Domain	Skill Level	Format	Score
1	BD3310	Clinical Educator - BD Pyxis ES System v1.5	Medication Management	Intermediate	E-learning	**0.6504**
2	BD1425	BD Rowa Vmax/Smart - Aufgabenplanung - DE	Medication Management	Intermediate	E-learning	**0.6362**
3	BD9772	CLP-ES300-PH: Pharmacy workflows Pyxis Anesthesia	Medication Management	Intermediate	E-learning	**0.6444**
4	BD7993	BD Pyxis ES v1.7.1 / CII Safe ES - LCR	Medication Management	Intermediate	E-learning	**0.6146**
5	BD1935	BD Pyxis MedStation ES Basics	Medication Management	Advanced	E-learning	**0.6239**

Score colour: green ≥ 0.65 (strong) | white = moderate | amber < 0.45 (weak)

TC-03 — UK ICU Nurse — TTM Expert Certification (No Contract, No OPCO)

VERDICT: PARTIAL PASS

Customer Profile

Customer ID	HCP0048
Role	Nurse
Specialty	ICU / Critical Care
Country	United Kingdom
Language	English (United Kingdom)
Skill Level	Intermediate
Establishment	Private Hospital
OPCO Eligible	No
Has BD Contract	No
Format Preference	E-learning
Available Hours	40 hours
Training Objective	Gain expert-level certification in TTM management
Preferred Product	BD Insyte™

Pipeline Recommendations

#	ID	Module Title	Topic Domain	Skill Level	Format	Score
1	BD3315	TTM U Advanced Level	Urology & Critical Care	Advanced	E-learning	**0.8903**
2	BD5868	Welcome to Hemodynamic University	Urology & Critical Care	Beginner	Webinar	**0.6700**
3	BD3732	TTM U Geavanceerd Niveau	Urology & Critical Care	Advanced	E-learning	**0.8512**
4	BD4279	TTM U Beginnersniveau	Urology & Critical Care	Advanced	E-learning	**0.8514**
5	BD8403	TTM U Livello Intermedio	Urology & Critical Care	Advanced	E-learning	**0.8459**

Score colour: green ≥ 0.65 (strong) | white = moderate | amber < 0.45 (weak)

TC-04 — UK ICU Specialist — Vague / Generic Objective (Edge Case)

VERDICT: FRAGILE — EXPECTED

Customer Profile

Customer ID	HCP0198
Role	ICU Specialist
Specialty	Specimen Management
Country	United Kingdom
Language	English (United Kingdom)
Skill Level	Beginner
Establishment	GP Surgery
OPCO Eligible	No
Has BD Contract	No
Format Preference	E-learning
Available Hours	40 hours
Training Objective	Support career development with certified training (Qualiopi equivalent)
Preferred Product	Pyxis ES

Pipeline Recommendations

#	ID	Module Title	Topic Domain	Skill Level	Format	Score
1	BD5491	BD Alaris™ neXus GP and Alaris™ GP/VP Volumetric	Medication Management	Expert	On-site	**0.4949**
2	BD3343	Best Practices: Reducing Pre-analytical Errors	Specimen Collection & Diagnostics	Advanced	E-learning	**0.4479**
3	BD8030	BD Pyxis CIISafe Formulary Workbook with ES	Medication Management	Intermediate	E-learning	**0.4554**
4	BD5868	Welcome to Hemodynamic University	Urology & Critical Care	Beginner	Webinar	**0.3885**
5	BD3944	Pharmacy workflows for BD Pyxis ES System v1.5	Medication Management	Intermediate	On-site	**0.3801**

Score colour: green ≥ 0.65 (strong) | white = moderate | amber < 0.45 (weak)

TC-05 — Italian System Manager — Vacutainer Blood Collection (Expert Level)

VERDICT: PARTIAL PASS

Customer Profile

Customer ID	HCP0237
Role	System Manager
Specialty	Geriatrics
Country	Italy
Language	Italiano (it) - Italian
Skill Level	Expert
Establishment	Policlinico
OPCO Eligible	No
Has BD Contract	Yes
Format Preference	Webinar
Available Hours	4 hours
Training Objective	Master blood collection techniques with BD Vacutainer® systems
Preferred Product	BD Microtainer®

Pipeline Recommendations

#	ID	Module Title	Topic Domain	Skill Level	Format	Score
1	BD5516	BD Rowa™ Vmax 210 - Risoluzione dei problemi - IT	Medication Management	Expert	Webinar	**0.4807**
2	BD7451	BD Microtainer® e BD Microtainer® MAP microeprouvettes	Specimen Collection & Diagnostics	Intermediate	Webinar	**0.4691**
3	BD0591	BD Vacutainer® Eclipse™ blood collection needle (Dutch)	Specimen Collection & Diagnostics	Expert	Webinar	**0.4453**
4	BD5817	TTM U Livello Avanzato	Urology & Critical Care	Intermediate	Webinar	**0.3805**
5	BD5270	System Manager - Pyxis MedStation ES v1.5: Essentials	Medication Management	Expert	Webinar	**0.3738**

Score colour: green ≥ 0.65 (strong) | white = moderate | amber < 0.45 (weak)

Section 3 — Test Case Summary

ID	Customer	Objective	Verdict	Top Score	Main Finding
TC-01	HCP0079	Développer les compétences en pose et en…	**PASS**	**0.6350**	Développer les compétences en pose et entretien de PICC line…
TC-02	HCP0077	Optimise Pyxis ES dispensing workflows…	**PASS — BEST CASE**	**0.6504**	Optimise Pyxis ES dispensing workflows…
TC-03	HCP0048	Gain expert-level certification in TTM m…	**PARTIAL PASS**	**0.8903**	Gain expert-level certification in TTM management…
TC-04	HCP0198	Support career development with certifie…	**FRAGILE — EXPECTED**	**0.4949**	Support career development with certified training (Qualiopi…
TC-05	HCP0237	Master blood collection techniques with …	**PARTIAL PASS**	**0.4807**	Master blood collection techniques with BD Vacutainer® syste…

Analysis & Next Steps

The five test cases above cover a range of profile types from highly specific clinical objectives to vague career development goals and collectively show that the pipeline produces directionally correct recommendations in the majority of cases. However, all testing so far has been conducted on synthetic data. Once the pipeline is connected to the real BD Academy dataset provided by Thomas, a proper evaluation pass will be needed: measuring whether the recommended modules align with actual training histories, completion records, and expert judgement from the BD business units. That evaluation will be the primary basis for parameter tuning specifically the fusion weights (currently 35% TF-IDF / 65% SBERT), the magnitude of each rule-based boost, and the size of the MMR candidate pool. Small adjustments to these values are expected to meaningfully improve recommendation quality once grounded in real usage data.